OSINT Frameworks and Metadata Forensics: A Technical Approach to Domain Auditing

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I. Objectives

The objective of this research is to demonstrate the efficacy of Open Source Intelligence (OSINT) in identifying organizational vulnerabilities. This includes:

- **Data Breach Correlation:** Linking specific domain assets to historical credential leaks.
- **Surface Mapping:** Enumerating subdomains and hosts to define the attack surface.
- **Forensic Artifact Analysis:** Extracting non-visible metadata from media files.
- **Search Engine Manipulation:** Leveraging Google Dorks to isolate sensitive directory structures.

II. Information Gathering and Email Leakage

A. The Breach Ecosystem

The investigation utilized aggregation platforms such as *HaveIBeenPwned* and *BreachDirectory*. These sites index data from "combo lists" and SQL injections. In this audit, **two major data breaches** were identified, confirming that credentials associated with the target have been exposed in the past, posing a risk of credential stuffing attacks.

B. Evolution of Harvesting Techniques

The tool `emailharvester` was tested but proved obsolete due to modern server-side protections. The research transitioned to **theHarvester**, which utilizes passive sources like search engine caches and active sources like DNS bruteforcing.

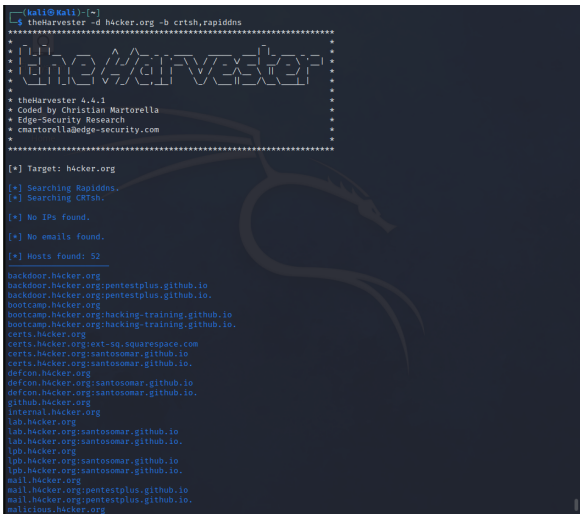

Fig. 1. Host enumeration via crtsh and rapiddns. 52 hosts identified.

As shown in **Fig. 1**, the tool successfully mapped the infrastructure. However, a broader search (**Fig. 2**) revealed a critical limitation: modern email scraping is heavily mitigated by web firewalls.

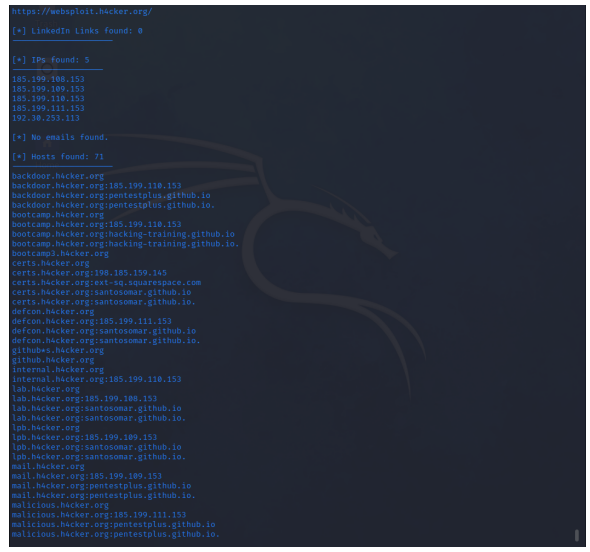

Fig. 2. Exhaustive search results showing IPs and links but no direct emails.

C. Advanced Reconnaissance with Spiderfoot

To automate the ingestion of APIs (Ahmia, Dehashed, Bing), the **Spiderfoot GUI** was deployed at `127.0.0.1:5001`. This allows for multi-threaded scanning of deep-web indices and archival data (CommonCrawl, Archive.org).

The **Ahmia module** (Fig. 3 and 4) was used to check for mentions on the Tor network. The null result suggests that the domain `h4cker.org` is not currently being traded or discussed in prominent underground forums.

III. Metadata Forensics: Exiftool

Metadata is often an overlooked leak in OSINT. **Exiftool** allows for the extraction of EXIF, IPTC, and XMP tags. In a real-world scenario, this can reveal the GPS coordinates of an office or the internal username of the person who created a document.

By using the command: `exiftool -csv Caracas.jpeg > data.csv` we transform individual file data into a structured format (Fig. 6) for automated processing.

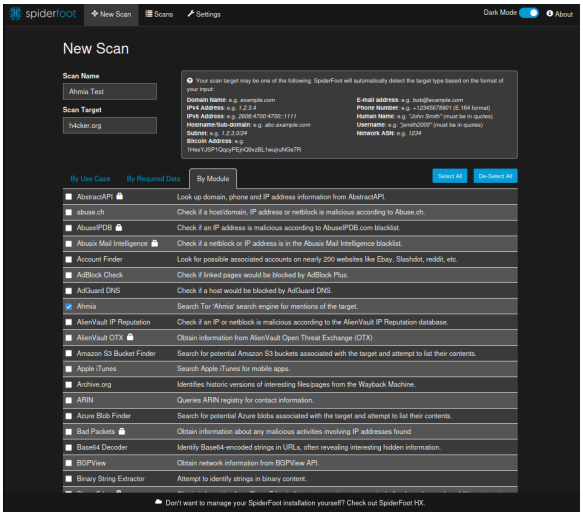

Fig. 3. Spiderfoot GUI: Configuring the Ahmia module for Dark Web indexing.

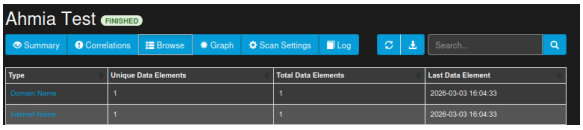

Fig. 4. Final Ahmia module report indicating zero .onion exposure.

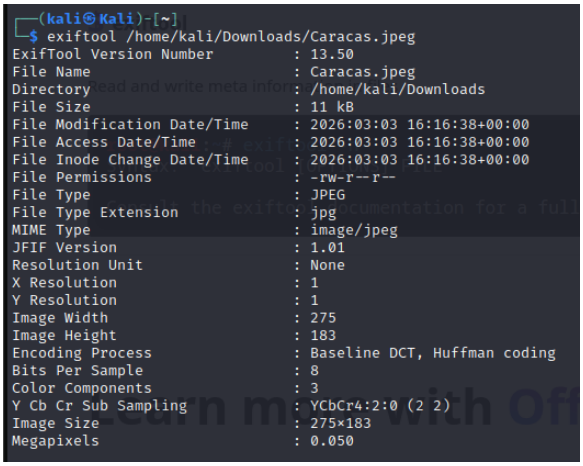

Fig. 5. Metadata extraction from Caracas.jpeg.

Fig. 6. CSV export showing structured metadata for forensic reporting.

IV. Advanced Google Dorking

Google Dorking utilizes specific operators to query the Google Hacking Database (GHDB). This allows auditors to find "low-hanging fruit" such as `phpinfo()` files or exposed `.env` files.

A. Key Operators and the GHDB

Operators like `intitle:`, `inurl:`, and `filetype:` allow for surgical precision. The **GHDB** serves as a repository of known-vulnerable queries. For instance, combining `site:target.com` with `filetype:sql` could reveal accidental database backups.

V. Conclusion

The shift from manual scraping to API-based tools like Spiderfoot is essential in modern OSINT. While tools like Exiftool provide micro-level data, Dorking and theHarvester provide the macro-level view of the organization's exposure.